Lavanya

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Summary

Final-year B.Tech. Computer Engineering student at Thapar Institute of Engineering & Technology with project experience in full-stack web development, REST APIs, and recommendation systems. Seeking entry-level roles in software and technology.

Education

Thapar Institute of Engineering & Technology — Patiala
Bachelor of Technology (B.Tech.), Computer Engineering — Expected May 2027 | CGPA: 8.07

Technical Skills

Languages: C++, Python, SQL, JavaScript
Frontend: HTML5, CSS3, Chart.js
Backend: Node.js, Express.js, FastAPI, REST APIs, JWT
AI & ML: Pandas, NumPy, Scikit-learn, EDA, Data Cleaning, Model Evaluation
Databases: MongoDB, MySQL
Core CS: Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN)
Tools: Git, GitHub, VS Code, Jupyter Notebook

Projects

Clario – Full-Stack Productivity & Analytics Platform
GitHub: https://github.com/lavu-create/Clario-FullStack-Productivity-System

- Developed a full-stack productivity SPA using Vanilla JavaScript, Node.js, Express.js, and MongoDB, with task/event management, mood tracking, and Chart.js analytics.
- Built a JWT-authenticated REST API with bcrypt password hashing and CORS controls; secured third-party weather integration through a server-side OpenWeatherMap proxy.
- Wrote 9 Jest/Supertest integration tests using mongodb-memory-server, covering authentication, protected routes, task CRUD, and CORS security; achieved 9/9 passing tests.

ShopSense – Multi-Tier E-Commerce Recommendation System
GitHub: https://github.com/lavu-create/ShopSense

- Processed 2.98M+ Amazon review interactions using Python, Scikit-learn, TF-IDF, and collaborative filtering to build a product recommendation system.
- Implemented Popularity, User-CF, Item-CF, and Weighted Hybrid models with cold-start routing, achieving 3.16% Recall@10 on a chronological test split of 39K+ interactions across 10K+ users.
- Deployed recommendations through FastAPI with lightweight routing optimized for a 512 MB hosting memory constraint.

Nexora – Assistive Communication Platform Prototype
GitHub: https://github.com/lavu-create/Nexora-Hackathon

- Developed a web-based prototype for an assistive communication platform for ICU-ventilated patients, with separate interfaces for patients, nurses, and family members.
- Implemented frontend workflows for communication, patient status, and monitoring using AI-assisted development tools.

Experience

Software Development Engineer Intern — Aug 2025 – Sep 2025
Bluestock Fintech — Remote

- Completed a remote internship with exposure to software development workflows in a fintech environment.

Achievements

- **Israel-India Hackathon 2025** – Advanced to Round 3 among 108 teams.
- **World Computer Hacker League (WCHL) 2025** – Advanced to the National Round.
- **GATE 2026 Qualified** – Officially qualified in the Data Science and Artificial Intelligence (DA) paper.

Extracurricular Activities

- **Performing Member, Mudra Dance Society, TIET** – Performed in university-level cultural events and competitions.

Languages

English, Hindi, Punjabi